



Real-Time User Risk Profiling in Financial Applications Using Unsupervised Learning on Streaming Data

- An Adaptive Framework for Risk Profiling in streaming Financial Data

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Traditional Fraud Detection is Reactive, Not Proactive.

- **Reliance on Static Data:** Traditional systems depend on rule-based engines and supervised models that require large, pre-labeled historical data
- **Slow to Adapt:** They struggle to keep pace with new and evolving threats because their training data quickly becomes outdated.
- **Lack of Personalization:** They often fail to understand an individual user's normal behavior, leading to a poor customer experience from false positives and a failure to detect novel attacks.



Shifting from Fraud Detection to Real-Time Risk Profiling

My Approach: I have designed, implemented and validated an end-to-end system that:

- **Learns Individually:** It learns each user's unique behavior using an unsupervised autoencoder, without needing pre-labeled data.
- **Adapts in Real-Time:** The system intelligently adapts to legitimate changes in a user's behavior using the ADWIN algorithm to handle concept drift.
- **Provides Dynamic Risk Scores:** It calculates a personalized, stateful Z-score for every transaction, offering a nuanced measure of risk.
- **Ensure Transparency:** It generates clear, feature-based explanations for its risk assessments using SHAP to build trust and meet regulatory requirements.

A Scalable, Low-Latency Streaming Architecture



- **Kafka:** The central messaging backbone that ingests high-throughput transaction data.
- **Spark Structured Streaming:** The powerful engine for stateful, real-time data processing and model application
- **PyTorch & SHAP:** The “brains” for the system, performing the unsupervised anomaly detection and generating explanations.
- **Streamlit Dashboard:** The real-time monitoring interface for visualizing performance, alerts, and explanations.

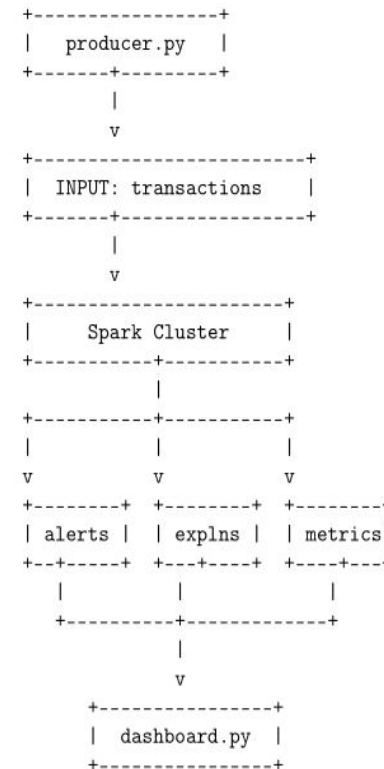


Figure 3.1: High-Level System Architecture

From Raw Data to a Personalized Risk Score

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- An incoming transaction is processed by the **PyTorch Autoencoder**, which calculates the *reconstruction_error*. A high error signals an unusual deviation from the user's normal pattern..
- The error is then normalized into a personalized, **Stateful Z-score**, which compares the current transaction's anomaly level to the user's recent history.
- If the Z-score exceed teh defined **ALERT_THRESHOLD** of 2.5, an alert is generated and sent to the monitoring dashboard.

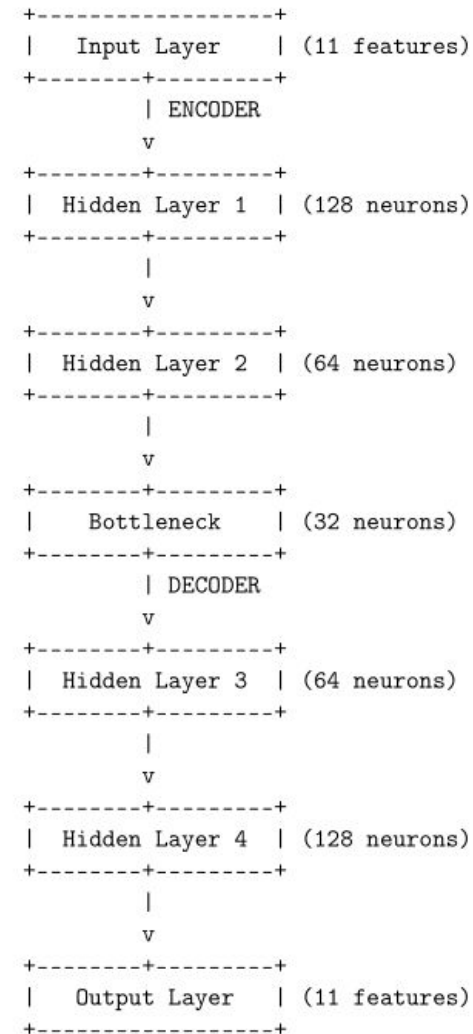


Figure 4.1: Autoencoder Model Architecture



Staying Relevant: How the System Adapts to Change

- **The Challenge:** A user's legitimate behavior can change over time. A static model would incorrectly flag these new, normal patterns as high-risk.
- **The Solution: ADWIN (ADaptive WINdowing)**
 - The system continuously monitors each user's stream of reconstruction errors.
 - When ADWIN detects a statistically significant change, it flags a drift event.
 - This automatically **resets the user's historical profile**, allowing the model to re-learn the "new normal" and preventing model decay.

Answering "Why?" with SHAP-based Explanations



- **Why it matters:**
 - **Regulatory Compliance:** Fulfills the “right to explanation” required by regulations like GDPR.
 - **Operational Trust:** Empowers human analysts to quickly understand and verify automated decisions.
- **How it Works (SHAP):**
 - For every high-risk alert, the system calculates the exact contribution of each transaction feature (like *amount* or *oldbalanceOrg*) to the final risk score.
 - This provides a clear, intuitive justification for why a specific transaction was flagged.

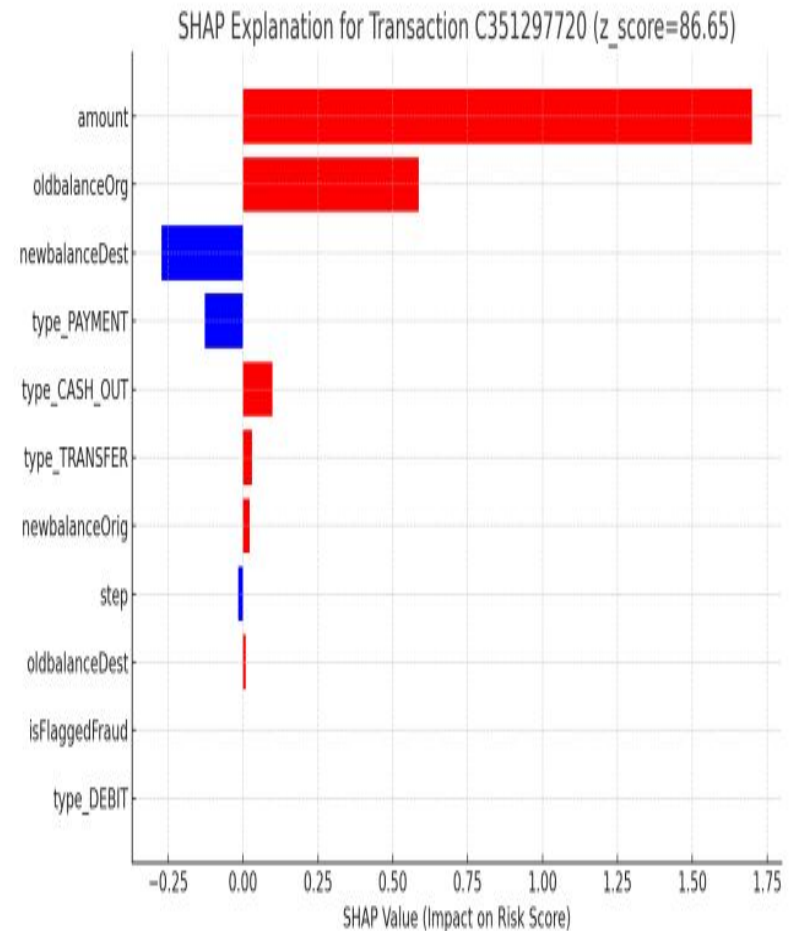


Figure 7.2: SHAP Explanation for a High-Risk Transaction

Stress-Tested and Validated for Real-World Throughput

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- **The Experiment:** I removed all artificial delays from the data producer to simulate a high-volume transaction flood and identify the system's operational limits
- **The Outcome:**
 - The system maintained a stable P95 latency **under 200ms** while processing up to approximately **1,200 transactions per second**.
 - This result confirms the architecture is robust and scalable for real-time financial monitoring applications.

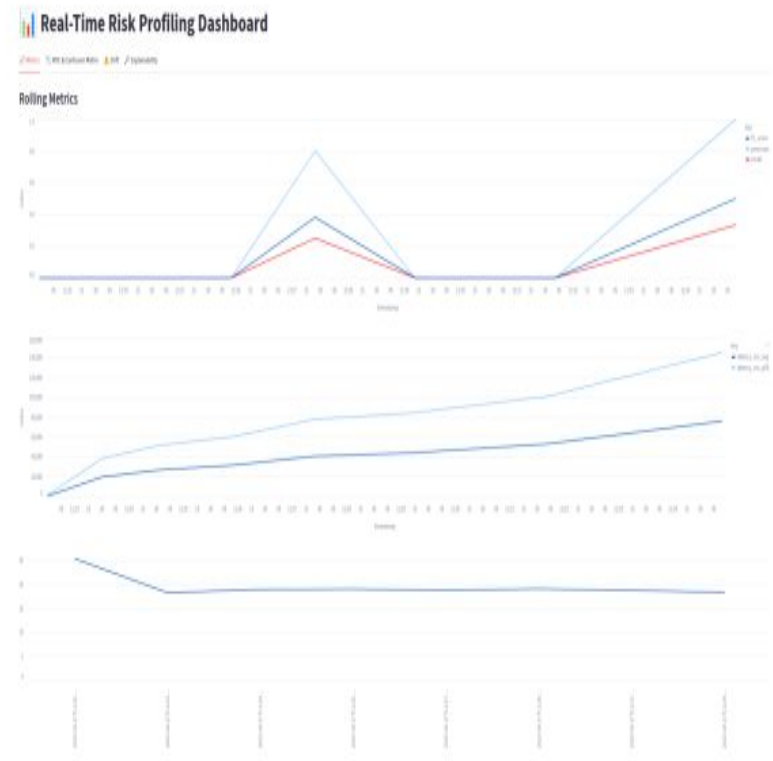


Figure 7.3: System Throughput vs. P95 Latency under stress conditions. The plot shows a performance inflection point where latency begins to increase exponentially as the system reaches its saturation point.

The System in Action: High Accuracy and Clear Insights

- **High-Risk Detection (True Positive):** The system correctly assigned a high Z-score of 86.65 to a known fraudulent transaction, with SHAP identifying *amount* and *oldbalanceOrg* as the key drivers.
- **Normal Behavior (True Negative):** It assigns a negligible Z-score of 0.42 to a routine payment, confirming its ability to ignore normal activity.
- **Overall Performance:** The confusion matrix shows a very low false positive rate, with the model correctly identifying over 10,947 legitimate transactions.

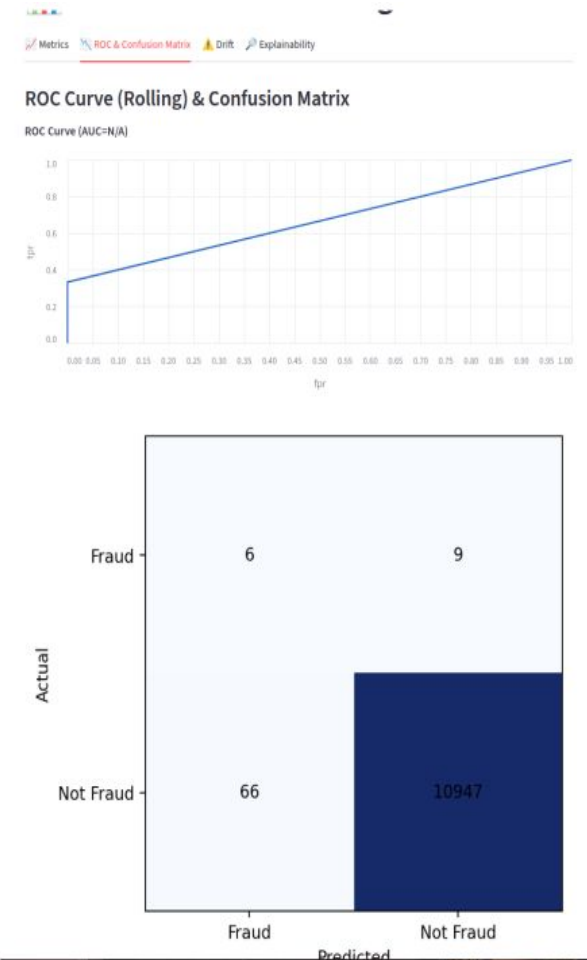
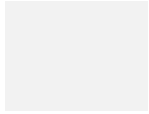


Figure 7.5: The ROC Curve and Confusion Matrix generated by the dashboard during a test run.



Live Demonstration



- **Summary of Achievements:**

- Successfully delivered a complete, end-to-end MVP for real-time, unsupervised risk profiling, meeting all 6 initial project objectives.
- The final system is **Scalable, Unsupervised, Adaptive, and Explainable**, addressing the core limitations of traditional systems.

- **Future Enhancements:**

- **Advanced Drift Handling:** Implement an online learning strategy to automatically fine-tune the autoencoder model after a drift event is detected.
- **Ensemble Modeling:** Improve accuracy further by integrating an ensemble of different unsupervised models (e.g. Isolation Forests, One-Class SVMs)

Q & A



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